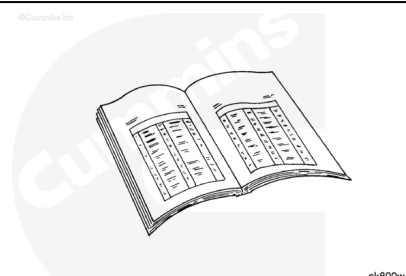


Trainings ▾

Preparatory Steps

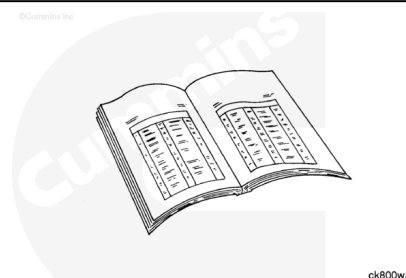
With Coolant Lines

- Remove the ECMs. See the Troubleshooting and Repair Manual Electronic Control System QSK50 and QSK60 Modular Common Rail System Series Engines, Bulletin 4021533. Refer to Procedure 019-031 in Section 19. (</qs3/pubsys2/xml/en/procedures/05/05-019-031.html>)
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html>)
- Remove the ECM cooling plate lines. Refer to Procedure 008-130 in Section 8. (</qs3/pubsys2/xml/en/procedures/56/56-008-130.html>)



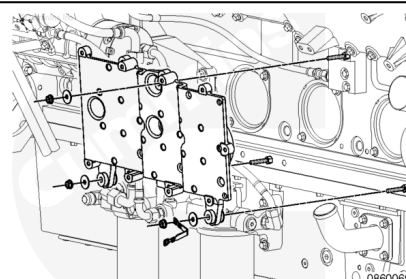
Without Coolant Lines

- Remove the ECMs. See the Troubleshooting and Repair Manual Electronic Control System QSK50 and QSK60 Modular Common Rail System Series Engines, Bulletin 4021533. Refer to Procedure 019-031 in Section 19. (</qs3/pubsys2/xml/en/procedures/05/05-019-031.html>)



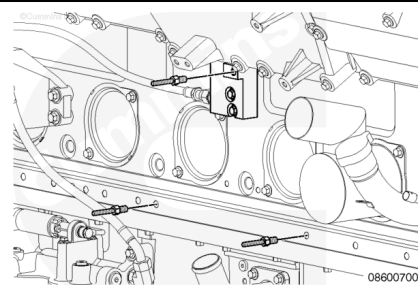
Remove

Remove the nuts and washers from the ECM cooling plate mounting studs and remove the ECM cooling plate.



Remove the ECM mounting studs.

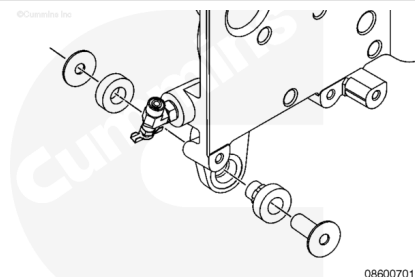
The upper stud is in the fuel junction block, with the bottom studs in the cylinder block.



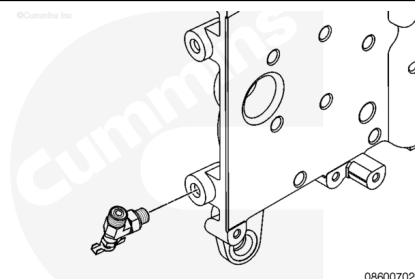
Disassemble

With Coolant Lines

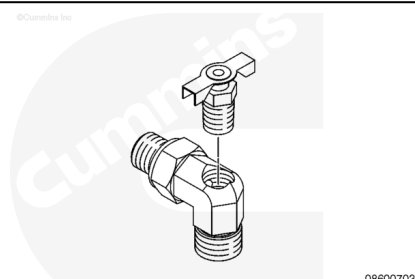
Remove the vibration isolators from the ECM cooling plate.



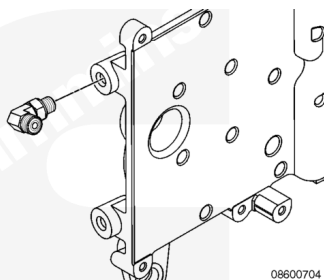
Remove the tee adapter from the ECM cooling plate coolant return port (lower).



Remove the drain cock from the return tee.



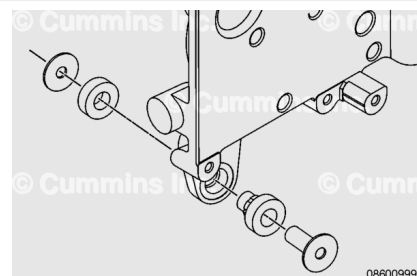
Remove the elbow adapter from the ECM cooling plate coolant supply port (upper).



08600704

Without Coolant Lines

Remove the vibration isolators from the ECM cooling plate.



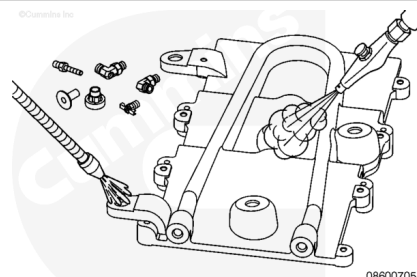
08600999

Clean and Inspect for Reuse

With Coolant Lines

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



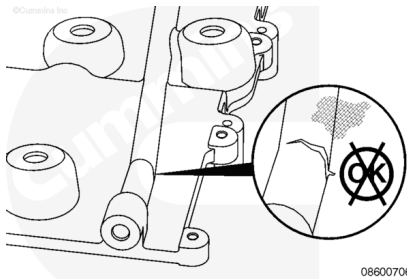
08600705

⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before use.

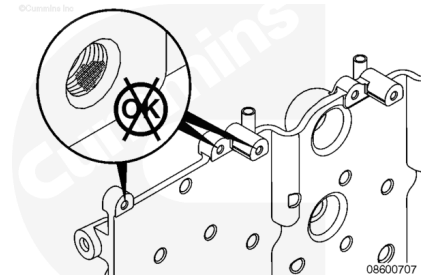
Clean the ECM cooling plate, fittings and attaching hardware with cleaning solvent, Cummins® Part Number 3824510 or equivalent.

Inspect the ECM cooling plate for any signs of cracks, corrosion, leaks, or other damage. Replace if damaged.



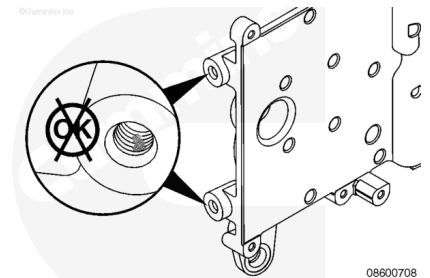
Inspect the threaded holes which mount the ECMs for any signs of damage.

If damage is found, it can be repaired with a screw thread insert, Cummins® Part Number 3822699 or equivalent (M8x1.25).

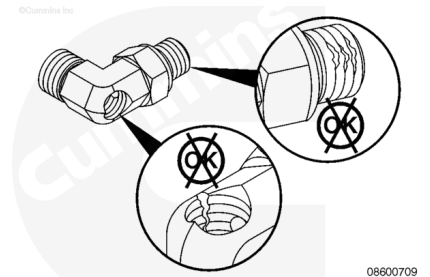


Inspect the threaded ports where the coolant lines attach for damage or stripped threads,

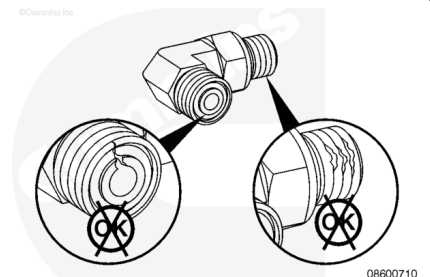
Replace the ECM cooling plate if the ports are damaged or the threads are stripped.



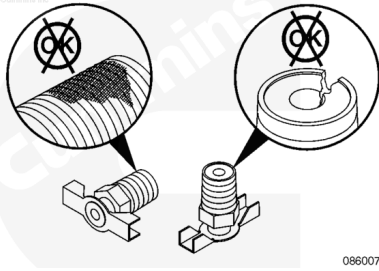
Inspect the tee fitting for signs of damage or stripped threads. Replace the tee fitting if damaged.



Inspect the elbow for cracks, stripped threads, or other damage. Replace the elbow if damaged.

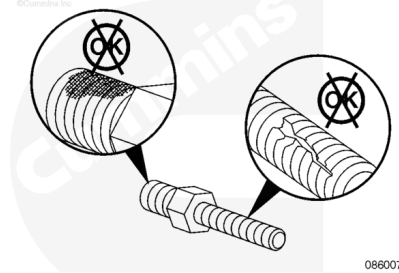


Inspect the drain cock for seizure, stripped threads, or other damage. Replace if damaged or seized.



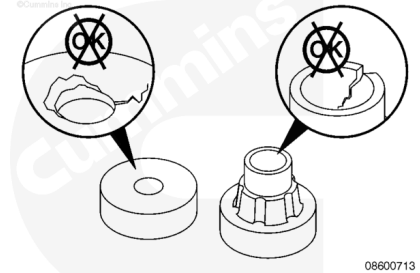
08600711

Inspect the ECM mounting studs for corrosion and stripped threads. Replace as necessary.



08600712

Inspect the vibration isolators for wear, cracks, or other damage. Replace if necessary.

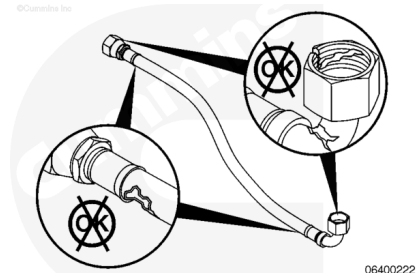


08600713

Inspect the ECM coolant supply and return lines for age, wear, cracks, leakage, or other damage.

Inspect the fitting on each end of the cooling lines for damage and stripped threads.

Replace damaged coolant lines.

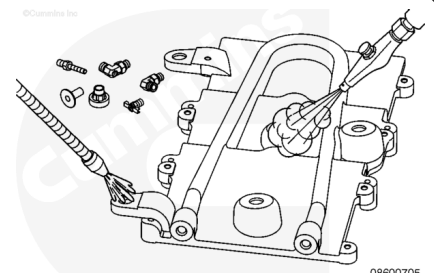


06400222

Without Coolant Lines

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



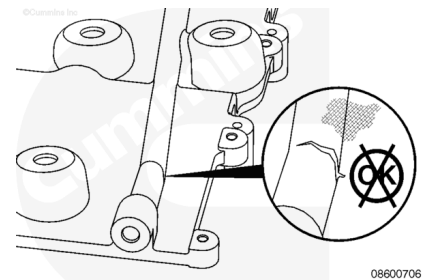
08600705

⚠ WARNING ⚠

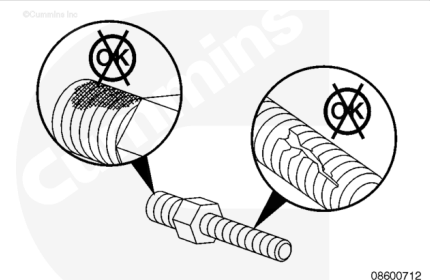
Some solvents are flammable and toxic. Read the manufacturer's instructions before use.

Clean the ECM cooling plate, fittings and attaching hardware with cleaning solvent, Cummins® Part Number 3824510 or equivalent.

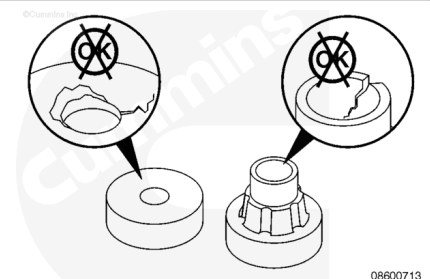
Inspect the ECM cooling plate for any signs of cracks, corrosion, leaks, or other damage. Replace if damaged.



Inspect the ECM mounting studs for corrosion and stripped threads. Replace as necessary.



Inspect the vibration isolators for wear, cracks, or other damage. Replace if necessary.

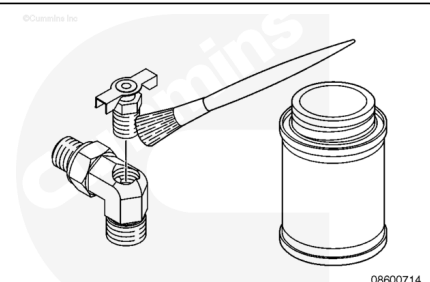


Assemble

With Coolant Lines

If reusing the drain cock, apply thread sealer, Part Number 3375066 or equivalent to the threads. If a new draincock is to be used, the threads will already have sealant applied.

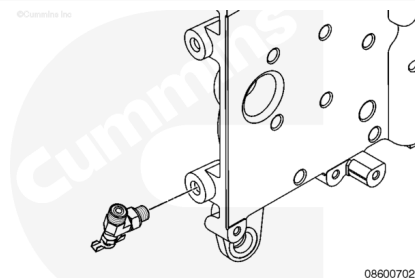
Install the drain cock into the tee. Hand tighten.



Install new o-rings on the return tee and install it in the ECM cooling plate return port.

Torque Value:

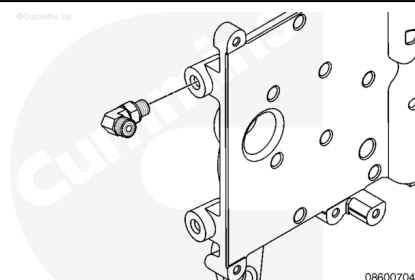
Coolant Return Tee 25 n•m [221 in-lb]



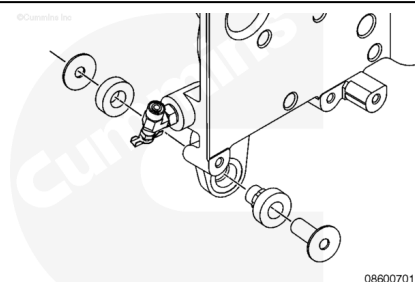
Install new o-rings on the supply elbow and install it in the ECM cooling plate supply port.

Torque Value:

Coolant Supply Elbow 25 n•m [221 in-lb]

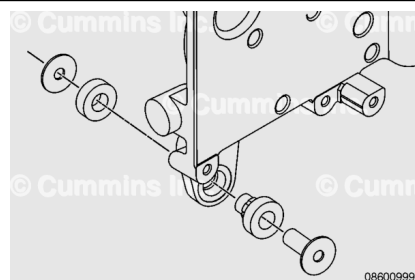


Install the vibration isolators on the ECM cooling plate.



Without Coolant Lines

Install the vibration isolators on the ECM cooling plate.

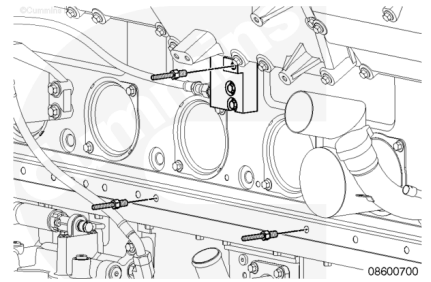


Install

Install the cooling plate mounting studs onto the fuel injection block and cylinder block.

Torque Value:

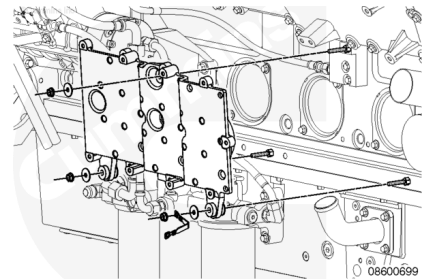
ECM Mounting Studs 80 n•m [60 ft-lb]



Install the ECM cooling plate and mounting nuts onto the mounting studs.

Torque Value:

ECM Cooling Plate Nuts 23 n•m [204 in-lb]



Finishing Steps

With Coolant Lines

- Install the ECMs. See the Troubleshooting and Repair Manual Electronic Control System QSK50 and QSK60 Modular Common Rail System Series Engines, Bulletin 4021533. Refer to Procedure 019-031 in Section 19. (</qs3/pubsys2/xml/en/procedures/05/05-019-031.html>)
- Install the ECM cooling plate lines. Refer to Procedure 008-130 in Section 8. (</qs3/pubsys2/xml/en/procedures/56/56-008-130.html>)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html>)
- Operate the engine and check for leaks.



Without Coolant Lines

- Install the ECMs. See the Troubleshooting and Repair Manual Electronic Control System QSK50 and QSK60 Modular Common Rail System Series Engines, Bulletin 4021533. Refer to Procedure 019-031 in Section 19. (</qs3/pubsys2/xml/en/procedures/05/05-019-031.html>)



Last Modified: 19-Aug-2009
